Supplement: Many Rows, Few Patients. Figures S1-S3.

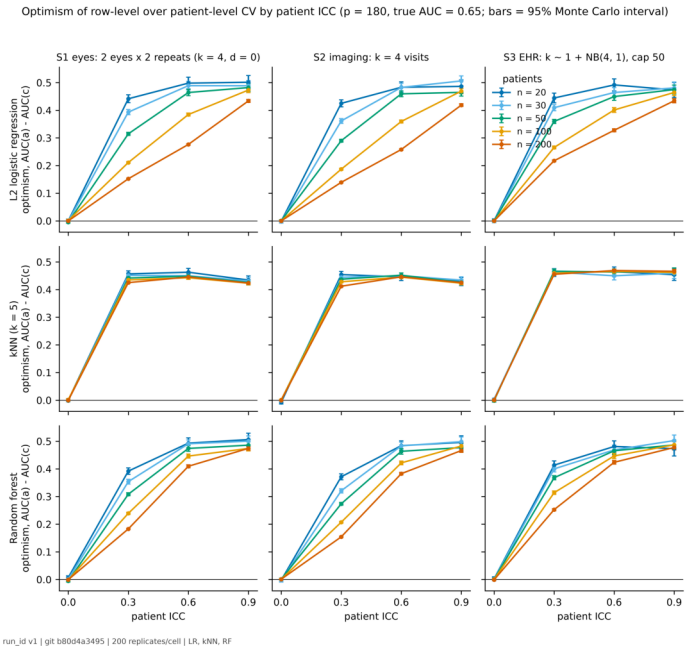


Figure S1. Optimism of row-level over patient-level cross-validation (difference in fold-mean AUC) against patient ICC, one line per number of patients. Columns: S1 (two eyes x two repeats, d = 0), S2 (four visits), S3 (EHR, rows per patient 1 + NB(4, 1)); rows: LR, kNN, RF; p = 180, true AUC 0.65; bars 95% Monte Carlo intervals; 200 replicates per point. (run_id v1, git b80d4a3495.)

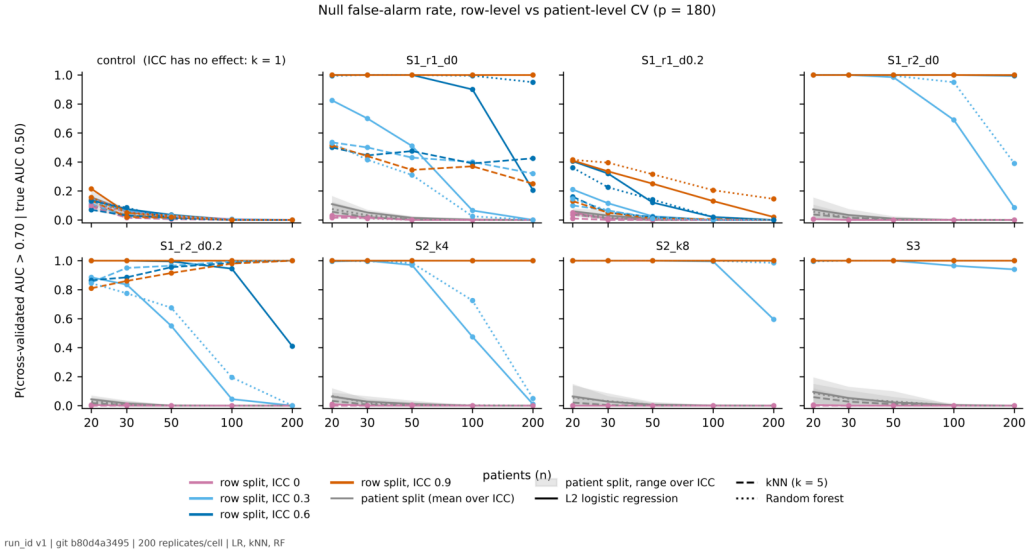


Figure S2. Probability that cross-validated AUC exceeded 0.70 when the true AUC was 0.50, against number of patients, by scenario variant. Coloured lines: row-level split, one colour per ICC; grey line and band: patient-level split, mean and range over ICC; line style: classifier; p = 180; 200 replicates per point. (run_id v1, git b80d4a3495.)

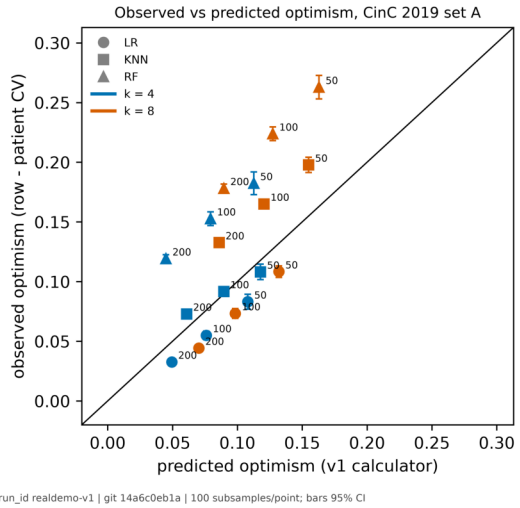


Figure S3. Observed optimism (row-level minus patient-level cross-validated AUC) against the calculator's prediction on the PhysioNet/CinC 2019 sepsis challenge training set A; k = 4 and 8 hourly rows per patient; labels give n; bars 95% CI over 100 class-balanced subsamples; line is identity. (run_id realdemo-v1, git 14a6c0eb1a.)

Repository: https://github.com/riziamm/data-leakage-study